Public EV Charging: Network Design and Financing

Executive Summary

This submission evaluates the city's shortlist for public charging infrastructure and recommends a financing structure for the first tranche of stations.

Analysis

An electric vehicle charging network is an infrastructure system of publicly accessible charging stations deployed along roadways and in urban areas to support battery electric vehicles.

Charging stations are typically categorised into three levels based on power output.

Level 1 charging uses a standard household outlet and delivers the slowest charge.

Level 2 charging uses dedicated equipment rated between 7 and 19 kilowatts and is the most common public option, while DC fast charging delivers 50 to 350 kilowatts and can recharge most vehicles in under an hour.

Network operators generally adopt one of two business models.

In the asset-owner model, the operator finances, builds and maintains the stations and earns revenue from electricity sales and usage fees.

In the franchise model, site hosts such as retailers or fuel stations own the hardware and pay the network operator for software, billing services and roaming access.

Revenue per station depends heavily on utilisation, and industry analysis suggests that most public stations require utilisation above fifteen percent to reach breakeven.

Grid integration is a central planning concern.

Unmanaged charging can coincide with evening demand peaks and force costly network upgrades, so utilities increasingly deploy smart-charging programmes that shift load to overnight periods or solar-rich hours.

Recommendation

We recommend the franchise model for the first sixty stations, with the asset-owner model reserved for three highway plazas where utilisation risk is lowest.

Case Prompt (identical for all candidates - do not modify): You are a strategy consultant engaged by the City Development Authority. Rising demand on public services and tightening municipal budgets have created pressure to identify fundable, high-impact interventions. Your task is to evaluate the feasibility of two shortlisted infrastructure initiatives, quantify expected costs and benefits over a five-year horizon, and recommend a phased implementation plan. Your submission must address financing options, stakeholder resistance, and delivery risks, and must include an executive summary, a structured analysis, and a clearly justified final recommendation supported by evidence. Word limit: 600 words. Citations are permitted where they support the analysis. Late submissions are scored as incomplete, and all claims must be verifiable from the provided case pack data.